

Question\_5:

Code:

```
1  /*
2  Question_5:
3  Find the maximum beauty Nesamani can guarantee while
4  painting sections of the building.
5  */
6
7  #include <iostream>
8  #include <cstring>
9  using namespace std;
10
11  int n, a[105], dp[105][105][105];
12
13  int solve(int l, int r, int dl, int dr) {
14      if (l == dl && r == n - dr - 1)
15          return 0;
16      int &ans = dp[l][r][dl];
17      if (ans != -1)
18          return ans;
19
20      ans = 0;
21      if (l > dl) {
22          int x = a[l - 1], best = 1000000;
23          if (l - 2 >= dl)
24              best = min(best, solve(l - 1, r, dl + 1, dr));
25          if (r + 1 <= n - dr - 1)
26              best = min(best, solve(l - 1, r, dl, dr + 1));
27          if (best == 1000000) best = 0;
28          ans = max(ans, x + best);
29      }
30      if (r < n - dr - 1) {
31          int x = a[r + 1], best = 1000000;
32          if (l - 1 >= dl)
33              best = min(best, solve(l, r + 1, dl + 1, dr));
34          if (r + 2 <= n - dr - 1)
35              best = min(best, solve(l, r + 1, dl, dr + 1));
36          if (best == 1000000) best = 0;
37          ans = max(ans, x + best);
38      }
39  }
```

```
39     return ans;
40 }
41 int main() {
42     int T;
43     cin >> T;
44     while (T--) {
45         cin >> n;
46         string s;
47         cin >> s;
48         for (int i = 0; i < n; i++)
49             a[i] = s[i] - '0';
50         memset(dp, -1, sizeof(dp));
51         int ans = 0;
52         for (int i = 0; i < n; i++) {
53             int best = 1000000;
54
55             if (i > 0)
56                 best = min(best, solve(i, i, 1, 0));
57             if (i < n - 1)
58                 best = min(best, solve(i, i, 0, 1));
59             if (best == 1000000)
60                 best = 0;
61             ans = max(ans, a[i] + best);
62         }
63         cout << ans << endl;
64     }
65     return 0;
66 }
```

**Output:**

**Test Case\_1:**

```
4
4
1332
6
4
9583
14
3
616
7
10
1029384756
31
PS C:\Users\Prabin Yadav\
```

**Test Case\_2:**

```
4
3
332
6
3
583
13
4
1616
7
8
29384756
24
PS C:\Users\Prabin Yadav\
```